

MID-TERM PROJECT REPORT

SaaS / E-Commerce Customer Intelligence Platform

Cohort Retention, CLTV Analysis and Churn Prediction System

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1. Abstract

This project report presents the mid-term progress of the SaaS / E-Commerce Customer Intelligence Platform, developed as part of an internship programme. The platform is designed to address critical business challenges related to customer retention, revenue optimization, and data-driven decision-making for SaaS and e-commerce enterprises.

The system integrates advanced data analytics, machine learning algorithms, and business intelligence frameworks to deliver a comprehensive Customer Intelligence solution. It encompasses modules ranging from RFM segmentation, Customer Lifetime Value (CLTV) estimation, and cohort retention analysis, to predictive churn modelling, AI-driven marketing intelligence, and executive-level decision support.

The Olist E-Commerce Dataset — a real-world Brazilian transactional dataset — serves as the primary data source for analysis and model development. The outcome is a unified Customer 360 Platform that enables organizations to proactively manage customer relationships, minimize churn, and drive sustainable revenue growth.

2. Problem Statement

Customer acquisition is among the most resource-intensive activities for modern businesses. Despite significant investment in acquiring customers, many organizations struggle to understand whether customers remain engaged post-acquisition or gradually disengage over time.

The absence of visibility into customer retention behaviour, customer lifetime value, churn patterns, and future revenue trends frequently leads to poorly informed business decisions and diminished profitability. Specifically, businesses face the following challenges:

- Inability to identify and retain high-value customers before they churn
- Lack of predictive analytics for early churn detection and proactive intervention
- Insufficient understanding of revenue distribution across customer cohorts
- Limited capability to forecast future revenue for budgeting and planning
- Absence of automated, data-driven business intelligence for executive decision-making

This project aims to resolve these challenges by developing an integrated Customer Intelligence Platform that provides end-to-end visibility into customer behaviour, value, and risk, enabling organizations to make informed and strategic decisions.

3. Project Objectives

3.1 Primary Objectives

- Analyse customer purchasing behaviour using RFM (Recency, Frequency, Monetary) methodology
- Measure customer retention and loyalty through cohort-based analysis
- Calculate and segment customers by Customer Lifetime Value (CLTV)
- Identify customers at high risk of churning using predictive machine learning models
- Forecast future business revenue based on historical trends

- Generate actionable, AI-driven business insights and recommendations
- Support strategic business decision-making at the executive level
- Develop a unified Customer 360 Intelligence Platform

3.2 Technical Objectives

- Build and validate a Random Forest-based churn prediction model
- Implement K-Means clustering for unsupervised customer segmentation
- Design and develop an Anomaly Detection Engine for business risk monitoring
- Automate report generation in PDF format for communication and governance
- Create a marketing intelligence module for targeted campaign recommendations

4. Dataset Overview

4.1 Primary Dataset

The project utilises the Olist E-Commerce Dataset, a publicly available, real-world dataset comprising Brazilian e-commerce transactions. The dataset contains multiple interconnected business tables that collectively represent the end-to-end order fulfilment lifecycle.

4.2 Dataset Components

Data Table	Description
Customer Information	Unique customer identifiers, geolocation, and demographic data
Orders Data	Order IDs, timestamps, status, and delivery information
Product Information	Product IDs, categories, dimensions, and descriptions
Payment Details	Payment types, instalment counts, and transaction values
Customer Reviews	Review scores, comments, and response timestamps
Seller Information	Seller identifiers and geographical data

4.3 Feature Engineering

The individual source tables were merged and transformed into a unified master analytical dataset. Key engineered features include:

- Customer purchase recency, frequency, and monetary value (RFM metrics)
- Customer tenure and activity period calculations
- Revenue metrics at the customer and order level
- Churn flag based on customer inactivity thresholds
- Cohort identifiers based on first purchase date

5. Technology Stack

Category	Tools / Libraries	Purpose
Programming Language	Python 3.x	Core language for all analytics and modelling
Data Manipulation	Pandas, NumPy	Data wrangling, aggregation, and feature engineering
Data Visualisation	Matplotlib, Seaborn	Graphs, charts, and retention heatmaps
Machine Learning	Scikit-Learn	Churn prediction and customer clustering
ML Algorithm	Random Forest	Binary churn classification model
Clustering Algorithm	K-Means	Unsupervised customer segmentation
Development Environment	Google Colab, Jupyter	Interactive development and experimentation
Business Intelligence	Power BI	Dashboard and KPI visualisation
Deployment (Planned)	Streamlit	Interactive web application deployment

6. System Architecture & Workflow

The platform follows a structured, multi-stage analytical pipeline designed to transform raw transactional data into actionable intelligence. The workflow is outlined below:

Stage 1	Data Collection	Aggregation of multi-table e-commerce transactional data from the Olist dataset.
Stage 2	Data Cleaning & Preprocessing	Handling missing values, inconsistencies, date conversions, feature engineering, and master dataset creation.
Stage 3	RFM Analysis	Customer behavioural segmentation based on Recency, Frequency, and Monetary value.
Stage 4	CLTV Analysis	Estimation and ranking of customers based on their total lifetime revenue contribution.
Stage 5	Cohort Retention Analysis	Cohort-based retention tracking and stickiness evaluation over time.
Stage 6	Business Insights Analysis	KPI dashboarding across revenue, orders, customer growth, and geography.
Stage 7	Product Analysis	Revenue contribution and trend analysis by product category.
Stage 8	Customer Clustering	K-Means based unsupervised segmentation into behavioural clusters.

Stage 9	Churn Analysis	Identification of at-risk customers, revenue at risk, and attrition patterns.
Stage 10	Churn Prediction	Random Forest model for binary churn classification and risk scoring.
Stage 11	Revenue Forecasting	Historical trend analysis and future revenue projection.
Stage 12	AI Insight Generation	Automated conversion of analytical outputs to business recommendations.
Stage 13	Executive Intelligence	Management-level KPI scoring, strategic risk assessment, and AI recommendations.
Stage 14	Customer 360 Platform	Unified customer profile integrating all intelligence outputs.

7. Modules Description

The platform comprises 16 functional modules, each addressing a distinct dimension of customer intelligence. The following table provides a consolidated overview:

#	Module Name	Key Activities	Business Impact
M1	RFM Analysis	Recency, Frequency, Monetary segmentation into Loyal, Active, At-Risk, and Inactive groups	<i>Targeted retention and marketing strategies</i>
M2	CLTV Analysis	Revenue estimation per customer, value segmentation, and customer ranking	<i>Prioritization of high-value customer retention</i>
M3	Cohort Retention	Cohort creation, retention matrix, and stickiness evaluation	<i>Identification of disengagement periods</i>
M4	Business Insights	Revenue, order volume, customer growth, delivery, and geographic KPIs	<i>Management-level business visibility</i>
M5	Product Analysis	Category revenue, product rankings, and monthly sales trends	<i>Product portfolio optimization</i>
M6	Customer Clustering	K-Means clustering by spending, frequency, and activity	<i>Personalized marketing enablement</i>
M7	Churn Analysis	Inactivity patterns, churn risk groups, and revenue at risk	<i>Early identification of at-risk customers</i>
M8	Churn Prediction	Random Forest model, risk classification, and future churn alerts	<i>Proactive retention strategy deployment</i>
M9	Revenue Forecasting	Historical trend analysis and future revenue projections	<i>Budgeting and strategic planning support</i>
M10	AI Insight Generator	Auto-generation of retention, revenue, and growth recommendations	<i>Reduced manual interpretation effort</i>

M11	Customer Health Intelligence	Health Index, Risk Score, Persona, Lifecycle Stage, and Strategic Value	<i>Holistic customer quality assessment</i>
M12	AI Marketing Intelligence	Targeting, campaign recommendations, upsell, and conversion scoring	<i>Improved marketing ROI and engagement</i>
M13	Executive Intelligence	BI Score, Growth Index, Risk Assessment, and AI recommendations	<i>Executive-level decision support</i>
M14	Automated PDF Reporting	Auto-generated executive summaries and KPI reports	<i>Streamlined governance and communication</i>
M15	Anomaly Detection Engine	Customer and risk anomaly detection with executive alerts	<i>Early warning for business issues</i>
M16	Customer 360 Platform	Unified dataset integrating CLTV, health, churn, persona, and marketing	<i>Complete customer intelligence view</i>

8. Expected Outcomes

Upon full implementation, the Customer Intelligence Platform is expected to deliver the following measurable business outcomes:

- Improved Customer Retention — Proactive identification and engagement of at-risk customers before churn occurs
- Reduced Churn Rate — Deployment of targeted retention campaigns informed by predictive churn probabilities
- Increased Revenue — Prioritization of high-CLTV customers and identification of upsell opportunities
- Enhanced Customer Segmentation — Granular, behaviour-based segmentation enabling personalized engagement
- Data-Driven Decision Making — Replacement of intuition-based decisions with analytics-backed insights
- Personalized Marketing — Campaign recommendations tailored to individual customer profiles and lifecycle stages
- Executive-Level Intelligence — Real-time business health scoring and strategic risk assessment for leadership
- Operational Efficiency — Automated reporting reducing manual analytics effort and improving communication speed

9. Conclusion

This project has evolved significantly from its initial scope as a traditional customer retention and CLTV analysis tool into a comprehensive, multi-layered Customer Intelligence and Business Analytics Platform. The system now integrates sixteen functional modules spanning data analytics, predictive machine learning, marketing intelligence, executive decision support, anomaly detection, and automated reporting.

By leveraging the Olist E-Commerce Dataset, the platform demonstrates practical applicability to real-world SaaS and e-commerce environments. The combination of descriptive analytics, predictive modelling, and AI-generated recommendations positions the solution as a robust framework for organizations seeking to optimize customer relationships and drive sustainable growth.

The mid-term milestone reflects substantial progress in building a production-ready Customer Intelligence Platform. Remaining work includes deployment via Streamlit, integration of the Power BI dashboard, and final model validation. These components are on schedule and will be completed in the second phase of the internship.

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